Lab 06_Hypothesis Testing

1. Introduction to Hypothesis Testing (Detailed)

Hypothesis Testing is a statistical method used to make decisions about a population based on sample data.

In real-world situations, it is often not possible to study the entire population. Therefore, we collect a sample and use statistical methods to test assumptions.

It helps answer:

- Is the result real?
- Or is it just due to random chance?

```
            Observed Result
                  |
                  ▼
        ┌──────────────────┐
        │ Real Difference? │
        │ OR Just Random?  │
        └──────────────────┘
                  |
        ┌─────────┴────────┐
        ▼                  ▼
   Significant        Not Significant
 (Reject H₀)         (Accept H₀)
```

Role in Artificial Intelligence

In AI and Machine Learning, hypothesis testing is used to:

- Validate model improvements
- Compare algorithms
- Evaluate experimental results
- Ensure that observed changes are statistically significant and not random

For example, if a model accuracy increases from 85% to 90%, hypothesis testing determines whether this improvement is statistically meaningful.

---

2. Basic Terminology

2.1 Population

The entire group of interest.
Example: All students in a university.

2.2 Sample

A subset of the population used for analysis.
Example: 25 students selected from the university.

2.3 Hypotheses

Null Hypothesis (H_0)

The default assumption that there is no effect or no difference.

Example:
The average marks are equal to 70.

Alternative Hypothesis (H_1)

The statement that contradicts the null hypothesis.

Example:
The average marks are not equal to 70.

```
H₀ (Null Hypothesis)        H₁ (Alternative Hypothesis)
---------------------      ----------------------------
No change                  There is a change
No improvement             Improvement exists
Default assumption         What we want to prove
```

Claim: Average marks = 70

- H_0: $\mu = 70$
- H_1: $\mu \neq 70$

Meaning:

- H_0 → marks did not change
- H_1 → marks changed

Types of Alternative Hypothesis

Type	Form
Two-tailed	$\mu \neq$ value
Right-tailed	$\mu >$ value
Left-tailed	$\mu <$ value

2.4 Significance Level (α)

What is α?

α (alpha) is the threshold used to decide whether to reject the null hypothesis.

$$\alpha = 0.05 = 5\%$$

Meaning:

We allow a **5% chance of making a wrong decision** (Type I error).

Important Clarification

α is **NOT fixed**. It is chosen based on the problem.

Common Values

α Value	Meaning
0.10	Less strict
0.05	Standard
0.01	Very strict

2.5 p-value

The p-value represents the probability of obtaining results assuming that the null hypothesis is true.

Decision Rule

- If p-value ≤ α → Reject H_0
- If p-value > α → Fail to reject H_0

```
p-value scale:

0 ──────────────── 0.05 ──────────────── 1
│        Reject H₀        │   Accept H₀   │
│   (Significant)         │ (Not Significant)
```

Interpretation:

- Left side → Strong evidence
- Right side → Weak evidence

2.6 Test Statistic

A numerical value calculated from sample data used to make a decision.

Examples:

- Z-value
- T-value

2.7 Errors

Type I Error

Rejecting a true null hypothesis.

Type II Error

Failing to reject a false null hypothesis.

---

3. Types of Hypothesis Testing

3.1 Z-Test

- Used when sample size is large (n ≥ 30)

- Population variance is known

3.2 T-Test

- Used when sample size is small ($n < 30$)
- Population variance is unknown

Types:

- One-sample T-test
- Independent T-test
- Paired T-test

3.3 Chi-Square Test

Used for categorical data.

3.4 ANOVA

Used to compare means of three or more groups.

```
                Hypothesis Testing
                        │
     ┌──────────────┬──────────────┬──────────────┐
     ▼              ▼              ▼              ▼
   Z-Test         T-Test      Chi-Square       ANOVA
 (Large data) (Small data) (Categories) (3+ groups)
```

---

4. Mathematical Example (Step-by-Step)

Problem

- Population mean (μ) = 70
- Sample mean ($\bar{x}$) = 75
- Sample size (n) = 25

- Standard deviation (s) = 10
- Significance level (α) = 0.05

Step 1: Define Hypotheses

- H_0: μ = 70
- H_1: μ ≠ 70

Step 2: Select Test

Since:

- Sample size < 30
- Standard deviation is from sample

We use a **T-test**

Step 3: Formula

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

(x̄ - μ) → Difference (Observed - Expected)

s → Data variation

√n → Sample size effect

Full formula → Measures how strong the difference is

Step 4: Calculation

Substitute values:

t = (75 − 70) / (10 / √25)

√25 = 5

t = 5 / (10 / 5)

t = 5 / 2

t = 2.5

Step 5: Decision

- Critical value (α = 0.05, df = 24) ≈ 2.06
- Calculated t = 2.5

Significance level: **α = 0.05**
Degrees of freedom: **df = 24** (df=n−1)
Critical value: ≈ **2.06**

α = 0.05 means **5% total rejection area**

Since this is a **two-tailed test (μ ≠ value)**:

$$\text{Each tail} = \frac{0.05}{2} = 0.025$$

So:

- Left tail = 2.5%
- Right tail = 2.5%

Since:

2.5 > 2.06

Reject H_0

Where does 2.06 come from?

The value **2.06** comes from a **T-distribution table**.

Steps:

1. Go to **df = 24 row**
2. Find column for **α/2 = 0.025 (two-tailed)**
3. The value you get ≈ **2.064**

Rounded → **2.06**

2.06 comes from the t-table and represents the cutoff point where only 5% of random outcomes lie beyond it (for df = 24).

Critical values of *t* for two-tailed tests

Significance level (α)

Degrees of freedom *(df)*	.2	.15	.1	.05	.025	.01	.005	.001
1	3.078	4.165	6.314	12.706	25.452	63.657	127.321	636.619
2	1.886	2.282	2.920	4.303	6.205	9.925	14.089	31.599
3	1.638	1.924	2.353	3.182	4.177	5.841	7.453	12.924
4	1.533	1.778	2.132	2.776	3.495	4.604	5.598	8.610
5	1.476	1.699	2.015	2.571	3.163	4.032	4.773	6.869
6	1.440	1.650	1.943	2.447	2.969	3.707	4.317	5.959
7	1.415	1.617	1.895	2.365	2.841	3.499	4.029	5.408
8	1.397	1.592	1.860	2.306	2.752	3.355	3.833	5.041
9	1.383	1.574	1.833	2.262	2.685	3.250	3.690	4.781
10	1.372	1.559	1.812	2.228	2.634	3.169	3.581	4.587
11	1.363	1.548	1.796	2.201	2.593	3.106	3.497	4.437
12	1.356	1.538	1.782	2.179	2.560	3.055	3.428	4.318
13	1.350	1.530	1.771	2.160	2.533	3.012	3.372	4.221
14	1.345	1.523	1.761	2.145	2.510	2.977	3.326	4.140
15	1.341	1.517	1.753	2.131	2.490	2.947	3.286	4.073
16	1.337	1.512	1.746	2.120	2.473	2.921	3.252	4.015
17	1.333	1.508	1.740	2.110	2.458	2.898	3.222	3.965
18	1.330	1.504	1.734	2.101	2.445	2.878	3.197	3.922
19	1.328	1.500	1.729	2.093	2.433	2.861	3.174	3.883
20	1.325	1.497	1.725	2.086	2.423	2.845	3.153	3.850
21	1.323	1.494	1.721	2.080	2.414	2.831	3.135	3.819
22	1.321	1.492	1.717	2.074	2.405	2.819	3.119	3.792
23	1.319	1.489	1.714	2.069	2.398	2.807	3.104	3.768
24	1.318	1.487	1.711	2.064	2.391	2.797	3.091	3.745
25	1.316	1.485	1.708	2.060	2.385	2.787	3.078	3.725
26	1.315	1.483	1.706	2.056	2.379	2.779	3.067	3.707
27	1.314	1.482	1.703	2.052	2.373	2.771	3.057	3.690
28	1.313	1.480	1.701	2.048	2.368	2.763	3.047	3.674
29	1.311	1.479	1.699	2.045	2.364	2.756	3.038	3.659
30	1.310	1.477	1.697	2.042	2.360	2.750	3.030	3.646
40	1.303	1.468	1.684	2.021	2.329	2.704	2.971	3.551
50	1.299	1.462	1.676	2.009	2.311	2.678	2.937	3.496
60	1.296	1.458	1.671	2.000	2.299	2.660	2.915	3.460
70	1.294	1.456	1.667	1.994	2.291	2.648	2.899	3.435
80	1.292	1.453	1.664	1.990	2.284	2.639	2.887	3.416
100	1.290	1.451	1.660	1.984	2.276	2.626	2.871	3.390
1000	1.282	1.441	1.646	1.962	2.245	2.581	2.813	3.300
Infinite	1.282	1.440	1.645	1.960	2.241	2.576	2.807	3.291

Step 6: Conclusion

There is sufficient evidence that the average marks are significantly different from 70.

5. Python Practical (Detailed Explanation)

Tools Used

- Python
- SciPy

Program

```
from scipy import stats

# Step 1: Define sample data (marks of students)
data = [72, 75, 78, 74, 77, 73, 76, 79, 71, 75]

# Step 2: Define population mean (hypothesized value)
mu = 70

# Step 3: Perform one-sample t-test
t_stat, p_value = stats.ttest_1samp(data, mu)

# Step 4: Print results
print("T-Statistic:", t_stat)
print("P-value:", p_value)

# Step 5: Define significance level
alpha = 0.05

# Step 6: Decision making
if p_value <= alpha:
```

```
    print("Reject Null Hypothesis")
else:
    print("Fail to Reject Null Hypothesis")
```

Line-by-Line Explanation

Line 1

from scipy import stats

This imports the statistical functions from the SciPy library.

The stats module contains functions for hypothesis testing.

Line 4

data = [72, 75, 78, 74, 77, 73, 76, 79, 71, 75]

A list of sample data is created. These values represent observed marks of students.

Line 7

mu = 70

This defines the population mean (null hypothesis value). We are testing whether the sample mean differs from 70.

Line 10

t_stat, p_value = stats.ttest_1samp(data, mu)

This performs a one-sample T-test:

- data = sample observations
- mu = hypothesized population mean

It returns:

- t_stat = calculated test statistic

- p_value = probability used for decision making

Line 13–14

```
print("T-Statistic:", t_stat)
print("P-value:", p_value)
```

These lines display the calculated values so we can interpret the result.

Line 17

```
alpha = 0.05
```

Defines the significance level (5%).

Line 20–23

```
if p_value <= alpha:
    print("Reject Null Hypothesis")
else:
    print("Fail to Reject Null Hypothesis")
```

This is the decision logic:

- If p-value is small → result is significant → reject H_0
- Otherwise → result is not significant

---

6. Lab Tasks

Task 1 (Manual)

- Take sample data
- Define hypotheses
- Apply T-test manually
- Write conclusion

Task 2 (Programming)

- Implement T-test using Python
- Interpret p-value

Task 3 (AI Application)

- Compare two model accuracies
- Apply hypothesis testing
- Conclude whether improvement is significant